



NEW PRODUCTS

MAKE IN INDIA

COMPONENTS

Automotive LED light driver

TPS92643-Q1 from Texas Instruments is a high-performance automotive buck infrared LED driver that can independently modulate the LED current using PWM dimming or analogue techniques. It offers a wide 5.5V to 36V operating input range and 40V tolerance that supports load dump for 400ms. The driver features an adaptive on-time average current-mode control, which is based on the inductor valley current detection. The adaptive on-time control enables a near-constant switching frequency set between 100kHz and 2.2MHz. Closed-loop feedback employed by the LED driver enables better than $\pm 4\%$ accuracy over wide input, output, and ambient temperature. The driver is suitable for use in automotive headlights and front fog-light driver module.

Texas Instruments
<https://www.ti.com>

Waterproof tactile switch

PTS645V is a waterproof tactile switch from C&K. Suitable for use in harsh industrial environments, it comes with a small PCB footprint and is tested for an electrical life of over 100,000 cycles. The low-profile SMT switch is available in multiple models having different heights and actuation forces to suit the needs of a wide variety of users. It comes with IP67 certification, which makes it highly durable and suitable for both indoor and outdoor applications in harsh environments, such as factories and



machine shops.

C&K switches
<https://www.ckswitches.com>

High-voltage relays

PhotoMOS HE relay series from Panasonic delivers increased creepage and clearance, stable on-resistance over a lifetime, and unlimited switching cycles. The relays offer 5kV I/O isolation with increased clearance and creepage distance on the output side. Unlike the electromechanical relays, the AQV258H5 HE series is unaffected by vibration and offers stable on-resistance during its operation. It also offers a miniature 1-Form-A DIP5 packaging, making it particularly suitable for battery management systems, charging stations, and other high-voltage measurement and infrastructure control applications.



Panasonic
<https://na.industrial.panasonic.com>

5G ICs for low-cost solutions

Anokiwave, Inc. has launched three new variants of the 4th generation 5G ICs: AWMF-0221, AWMF-0224, and AWMF-0210. These can address multiple use cases in the 26GHz 5G band, which was recently auctioned in India. The modules offer high efficiency, linear output power, small form factor, making them suitable for developing low-cost 5G solutions. All ICs support 3GPP-compliant 5G radio specs and high modulation bandwidths. The ICs also offer a mmW active antenna innovator kit as a reference design and starting platform



to evaluate and accelerate mmW 5G antenna designs. The 5G ICs are suitable for applications such as 5G base stations, consumer premise equipment, repeaters, etc.

Anokiwave
<https://www.anokiwave.com/index.html>

Barometric pressure sensor

ND015A from Superior Sensor Technology is an advanced barometric pressure sensor suitable for indoor spaces that enhances the accuracy of GPS elevation measurements. It can accurately determine the elevation within 15cm of the actual height and also measure elevation above or below sea level. Integrating ND015A with a GPS receiver enhances its capabilities and allows the GPS-enabled system to precisely pinpoint the latitude, longitude, and elevation with minimal errors. The pressure sensor has a wide measurement range of 1034mbar to 10.3bar. It is ideal for a wide variety of applications that need to measure air and gas pressures against absolute zero pressure.



Superior Sensor
<https://www.superiorsensors.com>

Ethernet PHY devices

Microchip's META-DX2+ enables secure Ethernet PHY family with port aggregation for enterprise and cloud interconnect. META-DX Ethernet PHY devices support Ethernet rates from 1 to 800G. META-DX2+ 's configurable 1.6T datapath architecture is 2x in total gearbox capacity and hitless 2:1 protection switch mux modes enabled by its unique ShiftIO capability. It also integrates port aggregation to maintain the most compact footprint in the transition to

